

Seminar

Classes and methods

1. Classes.

- Why do we need them?
 - Abstraction, reusability, single encapsulated objects with interface.
- Methods and **this** pointer.
 - Methods are **functions** inside of a class' declaration. They all have access to the **this** pointer.
 - **this** is a **const pointer** referring to the object that's called the method.
 - **this** can also be a **const pointer to a const** too.
- Constructors
 - Methods called when an object of a specific class is being created.
 - Constructors don't have a return type.
 - Default, parameterized and copy constructors.
More on the copy constructors and destructors - in the next lesson.
- Access modifiers.
 - **public:**
Everything after this modifier is visible by the outside world.
 - **protected:** (*more on this modifier when we learn about **inheritance***)
Everything after this modifier is visible by the children of the class.
 - **private:**
Everything after this modifier is NOT visible by the outside world.
- Differences with structs.
 - In **C++** almost none.
Structs have public access modifier by default.
Classes - private by default.
 - In **C** structs **can't** have methods, static members, access modifiers and more.
- Selectors and Mutators (getters and setters).
 - Each **selector** is a method that returns the value of a data member.

In code:

```
<member_type> Get<Member_name>() const
{
    return <member_name>;
}
```

*For data members that are **arrays** the return type of the **selector** must be a **const** <type>*, thus we don't break the encapsulation of the class.*

- Each **mutator** is a method that **modifies** a data member ONLY in the way we intend to. (i.e. we must validate the given parameter).

In code:

```
void Set<Member_name>(<member_type> value)
{
    if (<validate the given value>) {
        <member_name> = value;
    }
}
```

3. Examples.

Rectangle.h

```
#pragma once

class Rectangle
{
public:
    Rectangle();
    Rectangle(double width,
              double height);
    double calcArea();

    void setWidth(double width);
    double getWidth();
    void setHeight(double height);
    double getHeight();

private:
    double width;
    double height;
```

Rectangle.cpp

```
#include "Rectangle.h"

Rectangle::Rectangle()
{
    width = 0;
    height = 0;
}

Rectangle::Rectangle(double width,
                    double height) : Rectangle()
{
    setWidth(width);
    setHeight(height);
}

double Rectangle::calcArea()
{
    return width * height;
```

```
};

// The setter and getter
// for height are similar
// to the setter and getter
// for width

void Rectangle::setWidth(
    double width) {
{
    if (width >= 0)
        this->width = width;
}

double Rectangle::getWidth()
{
    return width;
}
```

Source.cpp

```
#include <iostream>
#include "Rectangle.h"

int main()
{
    Rectangle rect(4, 6);           // ⇔ Rectangle rect{4, 6};
    std::cout << rect.calcArea();

    return 0;
}
```